

$(t_0 \sigma_0 Q (FS_0 \dots (\text{stack (frame } e_0 l) F \dots) FS_1 \dots)) \longrightarrow$ [↓base]
 $(t_1 \sigma_1 Q (FS_0 \dots (\text{stack (frame } e_1 l) F \dots) FS_1 \dots))$

where $(\text{not value?}[e]), (\sigma_1 e_1) = \Downarrow \text{base}[\sigma_0, e_0]$

$(t_0 \sigma_0 Q (FS_0 \dots (\text{stack (frame } E[(\text{async/lambda } (x \dots_1) e) v \dots_1] l) F \dots) FS_1 \dots)) \longrightarrow$ [async-app]
 $(t_1 \sigma_1 Q (FS_0 \dots (\text{stack (frame } e x_{\text{async}}) (\text{frame } E[(\text{task } x_{\text{async}})] l) F \dots) FS_1 \dots))$

[task-return]

$(t_0 \sigma_0 Q_0 (FS_0 \dots (\text{stack (frame } v l) F \dots) FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma_1 Q_1 (FS_0 \dots (\text{stack } F \dots) FS_1 \dots))$

where $(\text{async? } l), x_{\text{async}} = l$

[task-exn-trap]

$(t_0 \sigma_0 Q_0 (FS_0 \dots (\text{stack (frame (throw } v_{\text{err}}) l) F \dots) FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma_1 Q_1 (FS_0 \dots (\text{stack } F \dots) FS_1 \dots))$

where $(\text{async? } l), x_{\text{async}} = l, v_{\text{obj}} = \text{lookup}[\sigma_0, x_{\text{async}}]$

[task-run]

$(t_0 \sigma_0 (FS_0 \dots (\text{stack (frame } E[(\text{run (lambda () e))}] l) F \dots) FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma_1 Q_1 (FS_0 \dots (\text{stack (frame } E[(\text{task } x_{\text{async}})] l) F \dots) FS_1 \dots))$

[await-continue]

$(t_0 \sigma Q (FS_0 \dots (\text{stack (frame } E[(\text{await (task } x_{\text{async}})]) l) F \dots) FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma Q (FS_0 \dots (\text{stack (frame } E[v] l) F \dots) FS_1 \dots))$

where $(\text{done } v) = \text{task-state}[\text{lookup}[\sigma, x_{\text{async}}]]$

[await-failed]

$(t_0 \sigma Q (FS_0 \dots (\text{stack (frame } E[(\text{await (task } x_{\text{async}})]) l) F \dots) FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma Q (FS_0 \dots (\text{stack (frame } E[(\text{throw } v)] l) F \dots) FS_1 \dots))$

where $(\text{failed } v) = \text{task-state}[\text{lookup}[\sigma, x_{\text{async}}]]$

[await-suspend]

$(t_0 \sigma_0 Q (FS_0 \dots (\text{stack current-frame } F \dots) FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma_1 Q (FS_0 \dots (\text{stack } F \dots) FS_1 \dots))$

where $(\text{async? } l), v_{\text{obj}} = \text{lookup}[\sigma_0, x_{\text{async}}], (\text{pending } _ \dots) = \text{task-state}[v_{\text{obj}}], \text{current-frame} = (\text{frame } E[(\text{await (task } x_{\text{async}})]) l)$

$(t_0 \sigma_0 Q_0 (FS_0 \dots (\text{stack (frame } E[(\text{os/io natural } v)] l) F \dots) FS_1 \dots)) \longrightarrow$ [os/io]
 $(t_1 \sigma_1 Q_1 (FS_0 \dots (\text{stack (frame } E[(\text{task } x_{\text{async}})] l) F \dots) FS_1 \dots))$

$(t_0 \sigma_0 Q_0 (FS_0 \dots (\text{stack (frame } E[(\text{os/resolve (task } x_{\text{async}}) t_{\text{resolve}} v)] l) F \dots) FS_1 \dots)) \longrightarrow$ [os/resolve]
 $(t_1 \sigma_1 Q_1 (FS_0 \dots (\text{stack } F \dots) FS_1 \dots))$

where $(\geq t_0 t_{\text{resolve}}), v_{\text{obj}} = \text{lookup}[\sigma_0, x_{\text{async}}]$

[os/resolve-blocking]

$(t_0 \sigma Q (FS_0 \dots (\text{stack (frame } E[(\text{os/resolve } v_{\text{task}} t_{\text{resolve}} v)] l) F \dots) FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma Q (FS_0 \dots (\text{stack (frame } E[(\text{os/resolve } v_{\text{task}} t_{\text{resolve}} v)] l) F \dots) FS_1 \dots))$

where $(< t_0 t_{\text{resolve}})$

[dequeue]

$(t_0 \sigma Q_0 (FS_{\text{main}} FS_0 \dots (\text{stack } FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma Q_1 (FS_{\text{main}} FS_0 \dots (\text{stack } F_{\text{head}} FS_1 \dots))$

where $\text{all-busy?}[FS_0, \dots], (F_{\text{head}} Q_1) = \text{q-pop}[Q_0]$

[unblock]

$(t_0 \sigma Q ((\text{stack (frame } E[(\text{block (task } x_{\text{async}})]) l) FS_{\text{rest}} \dots)) \longrightarrow$
 $(t_1 \sigma Q ((\text{stack (frame } E[v] l) FS_{\text{rest}} \dots))$

where $(\text{sync? } l), (\text{done } v) = \text{task-state}[\text{lookup}[\sigma, x_{\text{async}}]]$